

Instructions. (100 points) Sections 12.1–14.3. You may use a calculator. Show all work for credit.

- (10^{pts}) **1.** Let $A = (1, 3, -2)$, $B = (4, 1, 2)$, and $C = (2, 5, 0)$.

Find the scalar projection of $\overrightarrow{AC}$ onto $\overrightarrow{AB}$.

- (12^{pts}) **2.** Consider the two lines:

$$L_1 : \quad x = 2 + t, \quad y = 1 - t, \quad z = 3t$$

$$L_2 : \quad x = 4 + 2s, \quad y = -1 - 2s, \quad z = 6s$$

Are the lines **parallel**, **intersecting**, or **skew**? Justify your answer completely:

- If they are parallel, determine whether they are the **same line** or **two distinct lines**.
- If they intersect, find the point of intersection.

(16^{pts}) **3.** Consider the surface $4x^2 - y^2 + 4z^2 = 16$.

(a) (*8 pts*) Find the family of traces in the most important direction and at least one trace in each of the other two coordinate directions. Use these to give a complete description of the surface.

(b) (*8 pts*) Carefully sketch the surface, labeling the intercepts and the axis of symmetry.

(16^{pts}) 4. Let $\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t, 3t \rangle$.

(a) (4 pts) Show that the curve lies on a circular cylinder and give its equation.

(b) (7 pts) Find the unit tangent vector $\mathbf{T}$ at $t = \pi/2$.

(c) (5 pts) Find parametric equations for the tangent line to the curve at $t = \pi/2$.

- (12^{pts}) **5.** Find a vector function $\mathbf{r}(t)$ that represents the curve of intersection of the cylinder $y^2 + z^2 = 4$ and the surface $x = yz$.

(12^{pts}) 6. Consider the limit:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x y^4}{x^4 + y^4}$$

Does this limit **exist** or **not exist**? Provide careful evidence to support your conclusion. If it exists, prove it. If it does not exist, show why.

(12^{pts}) **7.** For $g(r, s, t) = e^r \sin(st)$, find the mixed partial derivative g_{rst} .

(10^{pts}) **8.** The equation $xz + yz^2 = 6$ defines z implicitly as a function of x and y near the point $(1, 1, 2)$.

Find $\frac{\partial z}{\partial x}$ at $(1, 1, 2)$.